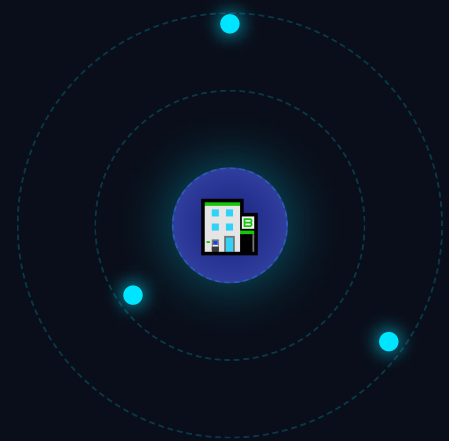


SBI SmartLife

The Autonomous Banking Growth Platform

Orchestrating customer lifetime value
through Customer Digital Twins and
Collaborative Multi-Agent AI.



The Passive Customer & Engagement Gap

- **Reactive Support:** RM advisory services are manually intensive and reserved for HNI clients.
- **Low Adoption:** Mobile banking features and digital products (YONO, UPI) suffer from low utilization rates.
- **Notification Blindness:** Generic marketing emails/SMS have click-through rates under 1.5%.
- **Missed Windows:** Loan pre-approvals are pitched **after** a customer buys a property.



The Living Digital Twin ()

- **Dynamic Modeling:** A 360° virtual representation of a customer's financial personality.
- **Behavioral Vectors:** Analyzes transaction velocity, category spend, and YONO login patterns.
- **Continuous Computation:** Evaluates Wellness, Credit, and Churn risk scores weekly.
- **Predictive Trajectory:** Evaluates what a customer will need 3 months before they request it.



Customer Digital Twin

Priya Patel • Active

Digital Maturity

78%

Wellness Score

72.5

Six Specialists, One Intelligent Graph

- **LangGraph Supervisor:** Intent classification and conditional edge routing.
- **Compliance Gate:** Built-in guardrails checking RBI guidelines and DPDP consent logs.
- **Specialists Ecosystem:** 6 specialized agents working in parallel.
- **Vector Database:** Semantic retrieval using ChromaDB for product descriptions and FAQs.

User Event Trigger



LangGraph Orchestrator (Supervisor)



6 Parallel Specialist Agents

Meet the AI Specialists

SLIDE 5

Specialist Agent	Core Responsibility	Hackathon Metric
Acquisition Agent	Conversational onboarding, smart KYC verification	Lead Conversion
Adoption Agent	Guides YONO, UPI, and digital features walkthroughs	Digital Maturity
Engagement Agent	Loyalty rewards, gamification, targeted campaigns	Click-Through Rate
Wellness Agent	Budget audits, savings milestones, debt coaching	Retention Rate
Life Events Agent	Transaction pattern logs to detect salary, marriage, housing	Cross-Sell Efficiency
Relationship Manager	Direct support, customer query routing & CSAT logs	CSAT Score

A Day in the Life of "Priya"

1

Salary Credit Detected

Priya receives a monthly transaction showing a 20% increase.

2

Twin Trait Updated

Life Events Agent detects the salary jump and updates the Twin profile.

3

Wellness & Engagement Match

Wellness Agent identifies investible surplus; Engagement Agent maps pre-approved SIP.

4

Conversational Activation

When Priya logs in, YONO Chatbot explains mutual fund advantages.

5

Complete Setup

SIP setup completed inside chat interface in under 60 seconds.

Enterprise-Grade Operator Dashboard

- **Customer Portal:** Personal AI chat, wellness score check, and interactive Twin.
- **Relationship Manager View:** Real-time customer lead logs, active campaigns, and event timelines.
- **Compliance Command Center:** DPDP Act consent monitors and RBI check logs.
- **System Administration:** Server health, API latency, and database connection pools.



Customer



RM Portal



Compliance



System Admin

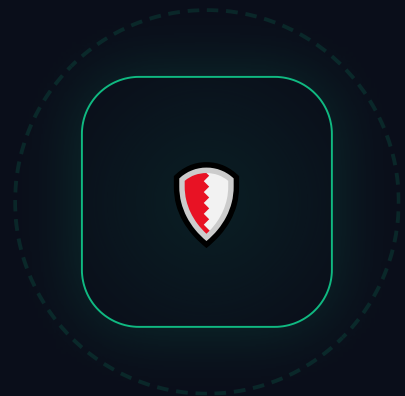
Turning Data into Retained Capital

- **Lower Acquisition Cost:** Automated conversational KYC and onboarding.
- **Enhanced Cross-Selling:** Targeted pre-approvals for loans and mutual funds.
- **Customer Retention:** High-value financial wellness coaching creates brand loyalty.
- **Branch Offloading:** 70%+ of basic queries handled instantly by specialized agents.

+18%Cross-sell
conversion**-40%**Onboarding
friction

Trust by Design, Compliant by Default

- **DPDP Act Consent Manager:** 1-click granular review and revocation of data access.
- **RBI Circular Guardrails:** Automated filters block misleading product recommendations.
- **Immutable Auditing:** All agent interactions are logged in a tamper-evident DB table.
- **Zero LLM Data Leakage:** Context filtering prevents PII from leaving private servers.



Async, Lightweight, & Containerized

- **Next.js 16 + React:** Interactive charts powered by Recharts, styled with Tailwind CSS.
- **FastAPI Backend:** High-performance asynchronous Python framework.
- **LangGraph + Gemini 2.5:** StateGraph execution with Gemini 2.5 Flash for high-speed reasoning.
- **ChromaDB + PostgreSQL:** Hybrid relational and vector storage.

FRONTEND

Next.js / Recharts

BACKEND

FastAPI (Async)

AI ENGINE

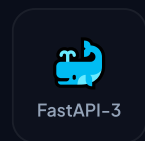
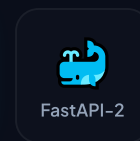
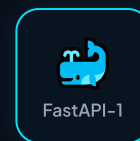
LangGraph / Gemini

DATABASE

PostgreSQL / Chroma

Ready for the Cloud, Built for On-Premises

- **Dockerized Setup:** Full stack deployment in one command via ``docker-compose``.
- **Scale-Out Ready:** Stateless FastAPI nodes behind load balancers.
- **Flexible Database:** Configured for local SQLite fallbacks and production PostgreSQL clusters.
- **Edge Deployment:** Compatible with private cloud environments for strict data sovereignty.



Enterprise Security & Data Safeguards

- **Data Encryption:** AES-256 encryption at rest for customer profiles and transaction records.
- **Secure Session JWT:** Encrypted tokens manage access to dashboard modules.
- **Strict Role-Based Access (RBAC):** Enforces clear separation between Customer, RM, Admin, and Compliance roles.
- **Rate Limiting:** Built-in middleware protects APIs from denial-of-service attempts.



Scaling to 500 Million Customers

Q3 2026

Core Banking System Integration

Real-time transaction streaming API connection.

Q4 2026

Branch RM Co-Pilot

Beta launch across 50 select flagship branches.

Q1 2027

YONO App Integration

Deployment directly inside YONO mobile app.

Q2 2027

Regional Languages

Multi-lingual voice banking assistance rollout.

Why SBI SmartLife Wins

- **Proactive vs. Reactive:** Doesn't wait for queries; automates financial wellness actions.
- **Unified Framework:** Combines 6 agents into a single collaborative LangGraph.
- **Compliance First:** Integrated DPDP consent managers and RBI filters.
- **Scalable Engine:** Async database pools and sub-second rendering.

Traditional Chatbots

- ✗ Static FAQ response
- ✗ Isolated chat logs
- ✗ No compliance filter

SBI SmartLife

- ✓ Proactive Twin Actions
- ✓ 6-Agent LangGraph
- ✓ DPDP Consent Built-in

Conclusion

Autonomous Banking is Here

Empower SBI's customers and relationship managers today.

State Bank of India Hackathon Team • Global Fintech Fest 2026

